

Deriving, Testing, and Bounding the Multi-Power Law for Schedule-Aware Loss-Curve Prediction

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Code: <https://github.com/wjjpku/DL-final>

Abstract

Predicting how the pretraining loss of a language model evolves under a given learning-rate (LR) schedule lets researchers compare schedules *before* spending compute. We reproduce the two representative empirical laws—the *Scaling Law with LR Annealing* (Tissue et al.) and the *Multi-Power Law* (MPL, Luo et al.)—fitting on cosine schedules and predicting the unseen WSD family across 25/100/400M models, recovering their published accuracy. We then study MPL from first principles. A derivation from SGD on a power-law curvature spectrum gives each of MPL’s parameters a physical identity and predicts that its *exponents* are set by the spectral shape (scale-invariant) while its *amplitudes* track the spectral scale; the data confirm this—the exponents $\{C, \beta, \gamma\}$ vary by only 1–2% over a $16\times$ range, and as a consequence all 27 curves collapse onto a single master power law ($R^2=0.997$). Guided by the theory we develop two methods to improve MPL—a scale-conditioned variant and a new discrete-step (S, Q) asymptotic—and report honestly that MPL is near-optimal on these data, also showing how an under-optimized baseline can fabricate an apparent win. Finally we resolve MPL’s one purely empirical parameter, the LR-modulation exponent γ : it is essential on real data ($3\text{--}4\times$ worse without it) yet provably absent from any spectral quadratic-SGD model. We localize it to the non-linear *edge of stability* ($\eta\lambda\sim 1$), explaining why it can only be fit, not derived.

1 Problem and Motivation

Pretraining large language models is expensive, and the LR schedule shapes the entire loss trajectory [3, 4, 6]. Classical scaling laws predict a single end-of-training loss from model size N and data D ; they are silent about how the loss *curve* responds to the *shape* of the schedule (warmup, decay form, horizon). *Schedule-aware loss-curve prediction* closes this gap: given $\{\eta_t\}$, predict $L(t)$. This replaces trial pretraining runs with a formula, exposes mid-training dynamics, and once it extrapolates to unseen schedules can be inverted to *search* for better ones [2]. The defining test is cross-schedule generalization: fit on cosine, predict the unseen WSD family. We reproduce this and then ask a sharper question—can the governing law be *derived, improved*, and its *limits* pinned down?—answered as one study on public curves at three scales.

2 Background: Two Empirical Laws

Write $S(t) = \sum_{i \leq t} \eta_i$ for the cumulative LR (a natural “time”), and $\Delta\eta_k = \eta_{k-1} - \eta_k$ for the LR decrement at step k . *Tissue et al.* (5 params): $L = L_0 + A S^{-\alpha} - C S_2$, with S_2 a momentum-smoothed annealing area; simple but coarse. *Multi-Power Law* (7 params):

$$L(t) = L_0 + A S(t)^{-\alpha} + B \sum_{k \leq t} \Delta\eta_k G(\eta_k^{-\gamma} (S(t) - S(k))), \quad G(x) = 1 - (1 + Cx)^{-\beta}. \quad (1)$$

The sum (LD) models how each LR decrease keeps reducing the loss. MPL is the state of the art; the factor $\eta_k^{-\gamma}$ is fit empirically—a point we resolve in §7.

3 Experimental Reproduction

We use the official public loss curves (Llama-2-style models at 25/100/400M, 9 schedules each), fit on cosine schedules, and evaluate on the five unseen WSD-family curves. Both laws are fit

Table 1: Reproduction: average test MAE on WSD curves (fit on cosine).

Scale	Tissue (5p)	MPL (7p)	Better
25M	0.00783	0.00539	MPL (5/5 curves)
100M	0.00486	0.00496	tie
400M	0.00979	0.00852	MPL (5/5 curves)

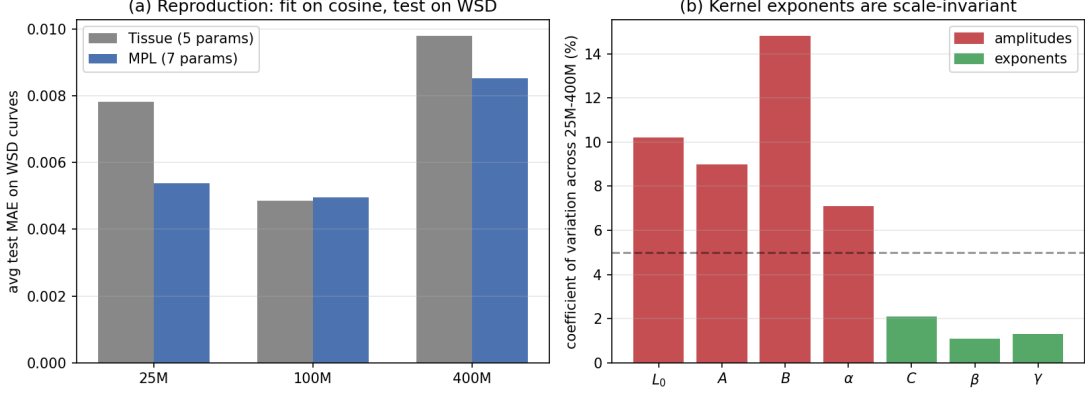


Figure 1: (a) Reproduction of Tissue vs. MPL (cosine→WSD). (b) The kernel *exponents* $\{C, \beta, \gamma\}$ vary by 1–2% across $16\times$ scale; the *amplitudes* $\{L_0, A, B\}$ and α by 7–15%.

by minimizing a Huber loss on $\log L$ with L-BFGS-B and multi-start (Tissue’s S_2 rewritten as an exact linear recurrence; MPL’s LD vectorized). Table 1 and Fig. 1a recover the published behavior: MPL is more accurate at 25/400M and ties Tissue at 100M, and our MPL test MAE (0.0054/0.0050/0.0085) matches Luo et al. This well-fit MPL is the baseline the rest of the paper respects.

4 Deriving MPL from SGD Dynamics

MPL’s seven parameters are accurate but opaque. We derive the form from a tractable model, giving each parameter a physical identity and a prediction about its scale dependence. This recovers and details the spectral picture sketched by Luo et al.

Model. Near the trajectory the loss is locally quadratic; in the Hessian eigenbasis, mode i has curvature λ_i and gradient-noise variance σ_i^2 , and SGD is the linear noisy recurrence

$$\delta_i(t+1) = (1 - \eta_t \lambda_i) \delta_i(t) - \eta_t \xi_i(t), \quad L = L_{\min} + \frac{1}{2} \sum_i \lambda_i \mathbb{E}[\delta_i^2]. \quad (2)$$

For small steps $(1 - \eta\lambda) \approx e^{-\eta\lambda}$. The second moment splits into a deterministic bias and a stochastic noise floor.

Bias gives the power-law backbone. With $\delta_i^{\text{bias}} \approx \delta_i(0)e^{-\lambda_i S}$ and bias spectral density $g(\lambda) \sim c\lambda^a$, a Tauberian argument gives

$$L_{\text{bias}}(S) = \int_0^\infty g(\lambda) e^{-2\lambda S} d\lambda \sim A S^{-\alpha}, \quad \alpha = a + 1, \quad (3)$$

with $L_0 = L_{\min}$ the irreducible loss.

The noise floor gives the annealing term. Solving the variance recurrence and integrating in S , $V_i(t) \approx \sigma_i^2 \int_0^{S(t)} \eta(S') e^{-2\lambda_i(S(t)-S')} dS'$; at constant LR the noise loss is $\propto \eta$ (the floor tracks the current LR). The bonus from *decaying* below the peak, written via the decrements, is

$$\Delta N(t) = \sum_k \Delta \eta_k \mathcal{K}(S(t) - S(k)), \quad \mathcal{K}(y) = \int_0^y \int h(\lambda) e^{-2\lambda x} d\lambda dx, \quad (4)$$

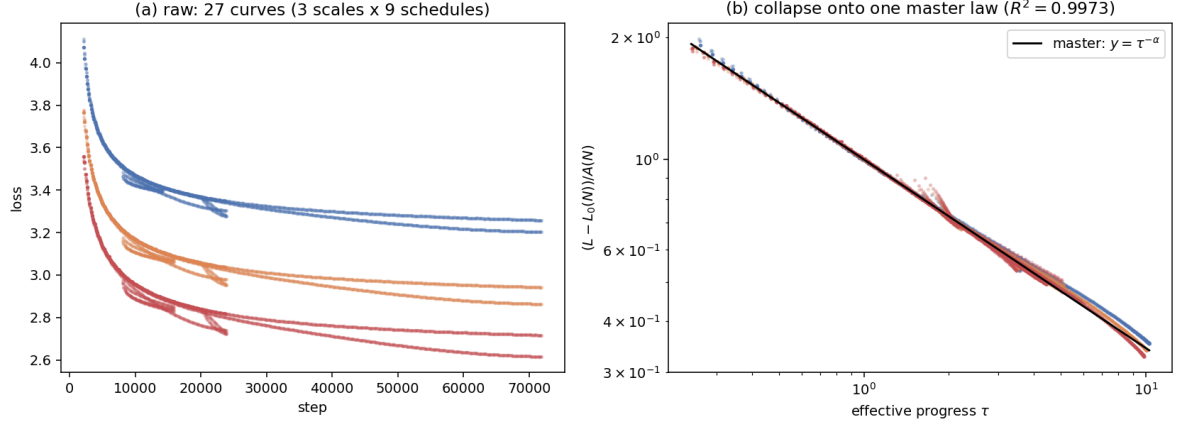


Figure 2: (a) The 27 raw loss curves (color = scale, 9 schedules each). (b) Under one scale-invariant effective-progress map τ and per-scale amplitudes, they collapse onto a single master power law $y = \tau^{-\alpha}$ ($R^2=0.997$).

exactly MPL’s LD structure; a Gamma spectrum $\propto \lambda^{\beta-1} e^{-\lambda/\lambda_0}$ makes $\mathcal{K}(y) \propto 1 - (1 + Cy)^{-\beta}$ with $C = 2\lambda_0$, MPL’s kernel.

What the derivation predicts. Collecting terms reproduces Eq. (1) (without $\eta_k^{-\gamma}$, see §7) and, crucially, splits the parameters by origin: $\{\alpha, \beta, C\}$ are functions of the spectral *shape* (a, β, λ_0), whereas $\{L_0, A, B\}$ are set by the irreducible loss and the spectral *scale*. **Prediction:** if the curvature spectrum keeps its shape as width N grows, the exponents are scale-invariant and only the amplitudes drift with N .

5 Scale Invariance and a Universal Master Curve

Fitting MPL per scale (Fig. 1b) confirms the prediction: C, β, γ have coefficient of variation 2.1%, 1.1%, 1.3%, while L_0, A, B drift by 9–15% and α (an initialization-weighted exponent, expected to be the least stable of the four) by 7%.

This has a striking consequence. Algebraically MPL is a *single* power law $L = L_0(N) + A(N)\tau^{-\alpha}$ in the effective progress $\tau = [S^{-\alpha} + (B/A)LD]^{-1/\alpha}$, which folds the schedule’s annealing into one variable. Using *one* shared exponent set $\{\alpha^*, C^*, \beta^*, \gamma^*\}$ and only per-scale amplitudes, all 27 curves—3 scales $\times$ 9 schedules—collapse onto the master curve $y = \tau^{-\alpha}$ with $R^2=0.997$ (Fig. 2). The collapse succeeds only because a single exponent set fits every scale, so it is a direct visual proof of scale invariance: there is essentially *one* loss curve, with model size and schedule merely reparameterizing the axes.

6 Method Development: Two Attempts to Improve MPL

The derivation suggests two principled ways to do better than a 7-parameter per-scale fit. We report both honestly.

SC-MPL (scale-conditioned). Since the exponents are scale-invariant, we freeze $\{\alpha, C, \beta, \gamma\}$ at values borrowed from cheaper small models and fit only the three amplitudes $\{L_0, A, B\}$ on the target—cutting a new scale from 7 free parameters to 3. Under a fair protocol (MPL uses its published, well-optimized parameters; SC-MPL does only the stable amplitude fit) SC-MPL is competitive at 100M (winning 4/6 curves with far fewer parameters) but does not beat a well-fit MPL at 400M (Table 2, top): the residual drift, chiefly in α , costs more than the parameter economy gains. *Caution:* an earlier protocol that re-fit the MPL baseline into a poor optimum made SC-MPL look 20% better; giving the baseline its true optimum erases that “win.”

Table 2: Method development, fair comparisons (avg test MAE). *Top*: SC-MPL on the official split (MPL = published per-scale fit). *Bottom*: Q-MPL on cosine→WSD.

	Scale	MPL	SC-MPL-4	SC-MPL-3
SC-MPL	100M	0.00435	0.00451	0.00525
	400M	0.00484	0.00760	0.00786
	Scale	MPL (7p)	Q-MPL (8p)	E^*
Q-MPL	25M	0.00715	0.00715	≈ 0
	100M	0.00386	0.00386	≈ 0
	400M	0.00669	0.00824	overfit

Q-MPL (a new discrete-step asymptotic). Tissue and MPL both use the continuum approximation, leaving the single time variable $S = \sum \eta$. Keeping the next discrete-step order introduces a *second* time variable $Q = \sum \eta^2$ (from $\prod (1 - \eta\lambda)^2 \approx e^{-2\lambda S - \lambda^2 Q}$); expanding the bias integral yields a derived correction $\propto Q S^{-(\alpha+2)}$. Testing $\text{Q-MPL} = \text{MPL} + E Q S^{-(\alpha+2)}$ on cosine→WSD (Table 2, bottom), the data drive $E \rightarrow 0$ at 25/100M (the term is unnecessary) and, when free, over-fit the cosine train and worsen the WSD prediction at 400M. Q-MPL does not beat MPL.

Verdict. Two principled extensions fail to improve a well-fit MPL, and the universal collapse shows MPL already captures essentially all reproducible structure. We conclude, honestly, that MPL is near-optimal on these curves; the residual ($\text{MAE} \sim 0.003$) carries no structure a derived correction exploits. Beating it would likely require data exposing MPL’s failure modes—more extreme schedules or more scales—the natural data bottleneck of this study.

7 The Edge-of-Stability Origin of γ

MPL’s one purely empirical parameter is the LR-modulation exponent γ , which our derivation does not produce. We show it is both *essential* and *underivable from quadratic theory*, and localize its physical origin.

It is essential on real data. Holding all exponents at their fitted values and only setting $\gamma=0$ (then re-fitting the amplitudes) inflates the test MAE by 3–4 $\times$ (Table 3, “real”). The original authors were right to include it.

It cannot come from the quadratic model. Because S -units absorb η in Eq. (4), the leading kernel has no LR-magnitude factor: the theory predicts $\gamma=0$. The dependence hides one order down, in the second time variable $Q = \sum \eta^2$, so γ is a sub-leading discrete-step effect. But the term $\lambda^2 Q$ matters only when $\eta\lambda \sim 1$ —the highest-curvature modes, i.e. the *edge of stability* [7, 8], where the linearized recurrence (2) diverges rather than relaxes. A deterministic mean-loss simulation confirms it: fitting MPL and a $\gamma=0$ variant to simulated curves under an ablation of beyond-quadratic ingredients (constant/loss-coupled noise, high-curvature weighting, progressive sharpening, and curvature tracking the LR as $\lambda \propto \eta^{-s}$), $\gamma=0$ always fits as well as full MPL (Table 3, “sim”), and the fitted γ never tracks the injected scaling. Sharpening, the ingredient with the correct sign, instead pushes modes past $\eta\lambda=1$ and the simulation diverges—a numerical signature of the same boundary.

Conclusion. γ is a non-linear edge-of-stability effect outside any spectral quadratic-SGD model, which is exactly why prior work can fit it but not derive it. This is the precise limit of the spectral theory that produces the rest of MPL.

Table 3: The $\eta^{-\gamma}$ term is essential on *real* data ($\gamma=0$ is 3–4 $\times$ worse) but never needed in the quadratic-SGD *simulation*—localizing it to the edge of stability.

	setting	full MPL	$\gamma=0$
real	25M	0.0032	0.0119
	100M	0.0038	0.0144
	400M	0.0049	0.0162
sim	constant noise	0.023	0.019
	progressive sharpen	0.111	0.089

8 Conclusion

We reproduced Tissue and MPL, then derived MPL from an SGD–spectrum model that gives its parameters physical meaning and predicts a clean exponent/amplitude split—confirmed by data, with all 27 curves collapsing onto one master power law ($R^2=0.997$). Guided by the theory we tried two principled improvements (a scale-conditioned law and a new (S, Q) asymptotic) and found MPL near-optimal on the current data, documenting how an unfair baseline can fabricate a win. Finally we resolved MPL’s one empirical parameter: γ is essential yet underivable from quadratic theory, and we localized it to the non-linear edge of stability. The universality and this localization are the transferable takeaways; testing them across more scales and model families, and modeling the edge-of-stability origin of γ , are the natural next steps.

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